RESEARCH PROTOTYPE / ENGLISH EDITION

Cross-Robot Real2Sim

Technical methods, measured evidence, failure analysis, and reproducible execution. Evidence freeze: 06 November 2026.

Task: **can pick, lift, transport, and place inside its assigned bin**. Source embodiment A is the 7-DoF Franka Panda; target embodiment B is the 6-DoF UR5e. Both operate in headless robosuite/MuJoCo PickPlaceCan.



Solid pipeline: executed bootstrap/fast. Human-video models remain an untested full-mode branch.

Executed experiment	Teacher placement	Saved learned policy	Feedback result
Bootstrap	3/3 source-retargeted UR5e	0/3 held-out seeds	1/3 on the same seeds
Fast	10/10 source-retargeted UR5e	9/10 held-out seeds	9/10; no improvement
Full / real human video	No real human clip supplied	Retargeting and training skipped	requires_human_video

Claim boundary

This is a working simulation infrastructure baseline using **oracle_sim_state**. Source information is limited to grasp offset and lift clearance inside a hand-coded controller. No real-human-video inference, vision-only policy, full-motion transfer, or physical-robot transfer has been demonstrated.

Original runs: bootstrap_20261105T181834Z and fast_20261105T182127Z. Later review videos re-execute frozen policies; they do not introduce new policy training.

Repository and experiment index

English case studies and provenance

01 / SYSTEM MAP

What the five stages establish

Each boundary has an independently runnable module and explicit evidence artifacts.

Boundary	Executed method / model	Input to output
1. Data Selection	Deterministic metadata filter and diverse-start selector; CLIP reserved for full mode.	Can HDF5 episodes to selection.json
2. Task Understanding	Pydantic-validated provided task seed; Qwen reserved for actual human frames.	Selection plus seed to task_spec.json
3. Cross-Robot Retargeting	Source-relative grasp/lift extraction plus BASIC/OSC_POSE feedback control.	Source trace plus task to executed UR5e NPZ
4. Policy Learning	26-128-128-4 MLP behavior cloning, fresh-process evaluation.	Successful executed teachers to checkpoint and eval.csv
5. Failure Analysis	Deterministic simulator-state rules and bounded additional data collection.	Actual rollout evidence to feedback and before/after metrics

Three separate meanings of success

Teacher success is placement achieved by the executed source-grounded controller. **Learned-policy success** is placement achieved by the saved MLP in closed loop on different simulator seeds. **Human-video understanding success** would require actual CLIP and Qwen inference on a real human clip; it is currently untested.

The dataset source is human-operated Panda robot demonstration data, not a video of an unaided person. The executable Panda scripted branch validates live source control, while the new original-state source videos reconstruct recorded HDF5 states. These are distinct evidence types.

1_Data Selection

2_Task Understanding

3_Cross-Robot-Retargeting

4_Policy Learning

5_Failure Analysis

02 / DATA AND ENVIRONMENT

Small, inspectable inputs

The implementation uses one minimal dataset file and records runtime compatibility rather than assuming checkpoint interchangeability.

Item	Verified value
Public dataset	robomimic/robomimic_datasets; v1.5/can/ph/low_dim_v15.hdf5; PH = proficient human robot demonstrations.
Downloaded scope	One file, 46,889,752 bytes, 200 recorded Panda episodes; no full repository download.
Dataset SHA-256	3f2eb92e0a5025d0095e866ac16cc8092d6a762abe27dec90dbaff9027282962
Recorded source environment	PickPlaceCan, robosuite 1.5.1 metadata, Panda, 20 Hz, OSC_POSE, lite_physics=false.
Executed simulator	robosuite 1.5.2; MuJoCo 3.3.7; offscreen EGL; 20 Hz; 300-step / 15-second episode cap.
Compute stack	Python 3.10.12; PyTorch 2.6.0+cu124; NumPy 1.26.4; h5py 3.14.0; Pydantic 2.12.3.
Host evidence	NVIDIA RTX 4090, driver 550.163.01; CUDA_VISIBLE_DEVICES=0. No heavyweight perception model used.

Contracts and units

JSON carries light metadata and provenance; NPZ carries typed trajectory arrays; CSV carries attempt/evaluation rows; MP4 supplies visual evidence. Positions are meters, times seconds, and angles radians. Named frames distinguish source_world, target_world, object_relative, and human_image_px.

The downloaded data include actions [T,7], EEF position [T,3], EEF quaternion [T,4], gripper observations, object state [T,14], simulator states [T,71], and an original model XML per episode. T varies by demonstration. No old learned robomimic checkpoint is loaded into the newer simulator.

Dataset inspection and recorded versions

Original execution environment

03 / STAGE 1

Data selection: what actually ran

Fast mode chooses five episodes from the first eight inspected source IDs; it does not search or rank the entire 200-episode corpus.

Deterministic selection algorithm

1. Enumerate source demos by numeric ID and inspect at most eight candidates (demo_0 through demo_7). Require nonempty actions and consistent observation lengths; retain the Can/Panda task provenance.
2. Sort coherent candidates by sample count, breaking ties by demo ID. Select the shortest demonstration first.
3. Repeatedly select the remaining candidate with the greatest minimum Euclidean initial-object-position distance to the already selected set. Stop after five selections.

```
selected = [shortest_consistent_demo]
while len(selected) < 5:
    next = argmax_d min_s ||object_start[d] - object_start[s]||_2
    selected.append(next)
```

Selection order	Samples	Initial can position [x,y,z] m
demo_3	98	[-0.00836, -0.08959, 0.86000]
demo_5	134	[-0.01880, -0.40074, 0.86000]
demo_7	115	[0.19077, -0.27073, 0.86000]
demo_4	102	[0.13630, -0.08713, 0.86000]
demo_1	118	[0.10021, -0.19378, 0.86000]

Output and limitation

selection.json records all inspected shapes, selected IDs, initial positions, source file hash, reasons, and strategy. The observed source-selection method is a metadata baseline. It supplies reproducible diversity but does not demonstrate task-aware learned retrieval or optimal source selection.

Human branch status
No human clip exists in the recorded runs. human_clip_selection is skipped_missing_input in bootstrap/fast. No CLIP score was computed, and no distractor rejection result is invented.

Selection evidence

04 / STAGES 1-2, FULL BRANCH

Human-video models: implemented, untested

Keep the proposed perception interface separate from the measured metadata-and-seed baseline.

Component	Implemented full-mode path	Current evidence
Video sampling	Scan data/human for real MP4 clips; sample 4-6 ordered frames at reduced resolution; preserve timestamps and annotated frames.	No human clip supplied.
CLIP ranker	openai/clip-vit-base-patch32 through Transformers CLIPModel / CLIPProcessor. Mean frame-to-task cosine similarity ranks actual clips.	Code exists; model inference not executed.
Qwen task interpreter	Qwen/Qwen2.5-VL-3B-Instruct through Qwen2_5_VLForConditionalGeneration, AutoProcessor, and qwen-vl-utils. Deterministic local generation on ordered images.	Code exists; weights and inference not tested in recorded runs.
GPU scheduling	CLIP, Qwen, and BC are sequential; model references are released between perception stages.	Bootstrap/fast load only the small BC model.
SAM2 segmentation	Optional interface reports the absence of implemented mask prompting/weights.	Not implemented; no segmentation achievement claimed.

Missing input is an auditable outcome

The recorded full run full_20261105T182257Z reports partial / requires_human_video. Step 2 has skipped_missing_input; retargeting and policy learning are skipped. Diagnosis records zero episodes and no feasible corrective iteration. A successful script exit or generated report does not imply human-video processing succeeded.

To validate this branch, supply a real clip with provenance and file hash; retain sampled frames, actual CLIP scores, Qwen prompt/raw text, parsed JSON, semantic mapping assumptions, and a human review of goal correctness. Then verify that the validated JSON gates the same downstream retargeting pipeline.

Recorded missing-human full report

CLIP implementation

Qwen implementation

05 / STAGE 2

Task understanding: provided semantics

Bootstrap and fast modes validate an explicit task specification; they do not infer goals from invented video.

Executed task representation

```
task_source: provided_seed
object: can
goal: {relation: inside, receptacle: bin}
goal_pose_source: simulator_oracle
unknowns: {mass: unknown, friction: unknown, contact_state: unknown}
```

The seed contains eight ordered stages: pre-grasp, grasp, close, lift, transport, lower, open, and retreat. Constraints require lifting before transport, clearing the source bin wall, and releasing over the assigned target compartment. Pydantic rejects unsupported object/goal relations and fabricated known physical properties.

Semantic label	Declared simulator binding / assumption
can	Can object in PickPlaceCan. The object identity is fixed for this MVP.
bin	The Can object's assigned target_bin_placements[can_index] entry. The visual receptacle in a future clip need not look identical.
Metric object and target poses	Read from privileged simulator state; no metric 3D position is reconstructed from monocular video.
Mass / friction / contact in task JSON	Remain unknown. Simulator contact used by controller/features is a separate privileged observation, not an inferred human-video contact fact.

Interface to retargeting

task_spec.json includes schema version, seed, mode, task source, model/controller name, input IDs, units, privilege flags, semantic mapping, and assumptions. Stage 3 validates the task before collection and refuses unsupported semantic mappings. The controller still implements only the fixed can-inside-bin scenario; it is not a general task planner.

Executed task JSON

Editable provided seed

Validated task schema

06 / STAGE 3, SOURCE GEOMETRY

What crosses the embodiment boundary

The source contribution is a measured grasp offset and lift clearance, not a copied joint trajectory or a learned source policy.

Schema discovery matters

For the actual 14D object observation, indices 0:7 contain EEF-relative pose; world can position is 7:10 and world quaternion is 10:14. Treating the first three values as world XYZ produces physically implausible geometry. The adapter explicitly decodes and tests this layout. EEF pose and actions are read from their named datasets. Timestamps are reconstructed from the recorded 20 Hz control rate.

Extraction and calibration

```
r[t] = eef_position[t] - can_world_position[t]
closing = {t : recorded_gripper_action[t] > 0}
grasp_t = argmin_t_in_closing ||r[t]||_2
grasp_offset = r[grasp_t]
source_lift = max_t can_z[t] - can_z[0]
```

When closing actions are absent, a geometric nearest-approach fallback is explicitly labeled. The target clips XY offsets to +/-25 mm and Z offset to [-15,40] mm. Lift is clipped to [0.12,0.25] m and combined with the configured minimum clearance. Object-relative axes remain world-aligned for the cylindrical can; this is a translation-relative construction, not a full rotating object-frame motion transfer.

Source demo	Applied grasp offset [x,y,z] mm	Applied clearance m
demo_3	[4.71, -2.26, 22.81]	0.20135
demo_5	[-10.33, 1.90, 14.24]	0.25000 (source lift 0.27388)
demo_7	[-11.30, 0.03, 8.64]	0.22358
demo_4	[-0.98, -0.09, 12.82]	0.21071
demo_1	[-0.54, 0.97, 24.12]	0.21320

Scientific limit / P02
The overall phase sequence, target generation, gripper timing, and orientation handling are hand-coded. Source quaternions may be stored, but their full time-varying orientation is not transferred. No source-parameter ablation has isolated the benefit of the source information.

Source extraction

Parameter calibration

07 / STAGE 3, EXECUTED CONTROL

The target uses its own closed-loop controller

Live UR5e feedback drives task-space waypoints; Panda joint angles are never copied into the 6-DoF target.

Phase	Target construction	Gripper / transition
Pregrasp	Current can + grasp offset + clearance in Z.	Open; EEF error below 28 mm advances.
Grasp	Current can + grasp offset.	Open; EEF error below 17 mm advances.
Close	Current can + grasp offset.	Close; 14 control steps.
Lift	Initial pickup position + grasp offset + clearance.	Close; EEF error below 28 mm.
Transport	Goal XY with initial object Z + offset + clearance.	Close; EEF error below 28 mm.
Lower	Goal XY with initial object Z + offset + 15 mm.	Close; EEF error below 17 mm.
Open	Same lower target.	Open; 14 control steps.
Retreat	Place position + clearance.	Open; evaluate actual placement predicate.

The teacher action is normalized XYZ = clip(0.6 * (waypoint - EEF) / 0.05, -1, 1), with positive gripper closure during close through lower. The phase tracker is computed online from simulator observations and elapsed phase steps. It is shared with policy evaluation, making this a hybrid learned-controller system rather than an autonomous learned phase planner.

Executed outcomes and retained provenance

Fast mode executed one scripted Panda success and ten source-retargeted UR5e successes. Bootstrap executed one scripted Panda success and three source-retargeted UR5e successes. The separate target_scripted_bootstrap fallback had zero attempts in these runs; therefore no source-versus-oracle comparison was performed. Attempt rows retain seed, source ID, origin, placement error, lift, and failure code.

Code does not prove physical infeasibility / P05
The historical unreachable_waypoint label means an early waypoint was not reached by the horizon. It is not an IK infeasibility certificate. At failed learned-policy seed 10000, the teacher succeeds from the same initial features.

Fast attempt-level CSV

Waypoint controller

08 / STAGE 4, EXECUTED DATA

What the target policy learns from

Only successful, actually executed UR5e teacher trajectories enter behavior cloning.

Source demos
3, 5, 7, 4, 1

UR5e teachers
10 executed successes

Shared B model
One baseline checkpoint

Held-out cases
10001 / 10003 / 10009

The fast baseline pools ten source-retargeted UR5e episodes from seeds 1042-1051. Five source IDs each contribute two executed target rollouts. These produce **1,620 state-action rows**; rows are temporally correlated samples from ten episodes, not 1,620 independent demonstrations. Bootstrap uses three episodes and 496 rows.

Feature group	Dimensions	Runtime source
EEF / can / goal XYZ	9	Actual target simulator positions.
Can minus EEF; goal minus can	6	Online coordinate differences.
Gripper opening, grasp contact, object lift	3	Observed gripper state, privileged sim contact, current Z minus initial Z.
Eight-phase one-hot encoding	8	Same online PhaseTracker in collection and inference.
Total	26	Runtime-validated, finite float32 vector.

Filtering and separation

The dataset builder accepts success=true, robot=UR5e, and teacher_origin in {source_retargeted, target_scripted_bootstrap}. Failed episodes, Panda episodes, learned-policy review rollouts, and any episode whose seed overlaps held-out evaluation are excluded or rejected. A zero-success dataset yields blocked_insufficient_teacher_data instead of invented training arrays.

Normalization mean and standard deviation are fitted only on accepted training rows; standard deviation has a 0.001 floor. The same statistics are saved with the model. Per-episode origins, source IDs, seeds, sample counts, and paths remain in dataset_metadata.json.

Known conditioning gap / P06
Source-specific grasp offset, lift clearance, and the current waypoint are not explicit components of the 26D input. They vary across teachers; evaluation uses the first successful teacher's tracker parameters. The structural mismatch is verified, while its contribution to the failure remains a hypothesis.

Fast training dataset metadata

09 / STAGE 4, MODEL AND LOSS

A small behavior-cloning model

The first learned target policy is deliberately low-dimensional; ACT, SmolVLA, and RGB policy learning are outside the executed baseline.

26D state
Training-only scaling

MLP 26-128-128-4
XYZ + gripper logit

4D intermediary
Clip XYZ; threshold grip

7D OSC vector
XYZ / zero rot / grip

Setting	Fast baseline value
Network	26 input features -> Linear 128 + ReLU -> Linear 128 + ReLU -> 4 outputs.
Outputs	Three normalized XYZ regression values and one gripper logit. XYZ is clipped; logit >= 0 closes, otherwise opens.
Optimizer / budget	Adam; learning rate 0.001; 2,000 updates; batch size 128; training seed 42; gradient norm clipped to 10.
Loss	L = MSE(predicted_XYZ, executed_XYZ) + 0.1 * BCEWithLogits(gripper_logit, close_label).
Class weighting	Positive-class BCE weight = (# open / # close), clamped to [0.25,4]; computed only from training rows.
Checkpoint	Model weights, model dimensions, optimizer, normalization, dataset provenance, step, CPU/CUDA/sampling RNG states.

Action adapter and units

The intermediary is not sent directly to robosuite. BASIC/OSC_POSE expects seven values: XYZ delta, three orientation deltas, and the gripper command. The adapter uses the installed composite controller's create_action_vector API, verifies 6D arm control and 7D environment action dimensions, inserts zero orientation deltas, and clips normalized values. Each XYZ unit maps to 0.05 m per control step in the tested controller.

Zero orientation delta holds the controller's initial orientation. BC does not learn orientation motion. Fresh-process evaluation loads the saved checkpoint and the same normalization and feature builder; training loss alone is never treated as task success.

Observed final losses: fast 0.000499 after 2,000 updates; bootstrap 0.002245 after 300 updates. These are fitting diagnostics, not independent control-performance measures.

Training implementation

Installed action adapter

10 / STAGE 4, HELD-OUT EVALUATION

Actual closed-loop target performance

One shared baseline checkpoint is evaluated on simulator seeds 10000-10009; these seeds are excluded from teacher training.

Seed	Native success	Steps / seconds	Can-to-goal XY error mm
10000	No	300 / 15.00	756.73
10001	Yes	150 / 7.50	8.48
10002	Yes	154 / 7.70	5.95
10003	Yes	156 / 7.80	4.86
10004	Yes	158 / 7.90	4.24
10005	Yes	151 / 7.55	9.08
10006	Yes	164 / 8.20	3.53
10007	Yes	154 / 7.70	6.87
10008	Yes	170 / 8.50	5.38
10009	Yes	225 / 11.25	10.33

What is measured

Each episode executes the learned XYZ/gripper outputs through the same action adapter and online phase tracker. The original goal predicate delegates to installed robosuite _check_success(). Mean placement error is the final can-to-target **XY distance**; it is not a full 3D pose/orientation score. Fast baseline success is 9/10; all-episode mean XY error is 81.55 mm, dominated by failed seed 10000. Success-only mean is 6.53 mm.

Stronger review predicate is separate / P09

Later review re-executes the frozen policy and checks the specific Can inside its assigned compartment, no grasp contact, and one additional second under the same policy. No scripted override assists the learned rollout. These stability annotations supplement rather than overwrite the original evaluation. Review videos and NPZ are excluded from training.

Evaluation limits / P07
Ten simulator seeds and one training seed do not establish broad generalization. The policy depends on exact simulator state and a manual phase tracker. No unseen object class, task family, camera estimator, physical robot, or perception robustness result is available.

Original fast evaluation CSV

Review stability annotations

11 / STAGE 5, DIAGNOSTIC RULES

Evidence first, causal claims second

Diagnostics operate on executed state/action arrays and native task outcomes; no large model performs failure attribution.

Recorded condition	Rule / evidence	Interpretation limit
Task success	Native simulator goal predicate; later review adds specific-Can, release, and stability checks.	Different predicates stay separate.
Lift achieved	Maximum can Z minus initial can Z >= 0.06 m.	This is measured motion, not proof of a secure grasp.
Object drop	Task failed, lift achieved, final object height below initial + 0.03 m.	A deterministic rule on simulator state.
Premature release	Close-to-open action change while lifted > 0.04 m and > 0.10 m from goal XY.	A plausible failure symptom, not unique cause.
Gripper/action mismatch	Wrong action dimensions, nonfinite values, or magnitude outside normalized bounds.	Adapter integrity check, not contact inference.
Grasp failed	Insufficient lift with no recorded contact, where the retained failure code supports this diagnosis.	Missing observations remain unknown.
Early waypoint / timeout	Retained phase/horizon code plus phase and EEF-object records.	unreachable_waypoint is not proven infeasibility.
Collision / semantic mismatch	Use explicit collision logs when present; unsupported parsing/selection causes remain unknown.	Do not force a physical or semantic cause.

failure_report.csv contains one row per analyzed episode with seed, teacher origin, step attribution, reason, XY error, lift/drop/release flags, diagnostic_oracle_used=true, confidence type, and evidence. Native task outcomes and deterministic rule-based interpretations are distinguished from unknowns.

feedback_requests.json ranks observed failure codes and suggests bounded experiments. The implemented automatic intervention is generic successful-teacher collection plus BC retraining. The failure-report language must not imply that a causal model selected a proven remedy.

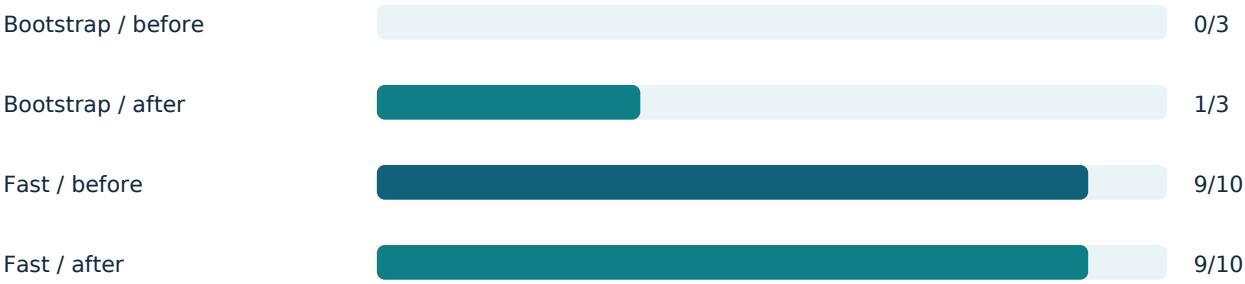
Detector implementation

Feedback request evidence

12 / STAGE 5, CORRECTIVE ITERATION

More data did not solve the fast failure

The before/after comparison uses identical evaluation seeds and retains a negative result.



Metric	Bootstrap before -> after	Fast before -> after
Held-out success	0/3 -> 1/3	9/10 -> 9/10
All-episode mean XY error	653.82 -> 258.88 mm	81.55 -> 84.58 mm
Additional teacher successes	2 from new training seeds	3 from seeds 542-544
Fast success-only mean error	Not the primary reported bootstrap statistic	6.53 -> 9.89 mm
Improvement claim	A measured small-set increase only	false; the same failure persists

Executed intervention

After observing learned-policy failures, select a feasible successful source-retargeted teacher origin, collect bounded additional trajectories on training seeds distinct from the evaluation set, merge successful data with the original teachers, retrain BC, and evaluate on the same held-out seeds. Baseline and corrective fast models each use 2,000 updates; bootstrap uses 300 updates. The intervention does not directly query or add the failed held-out trajectory to training.

The fast failure at seed 10000 remains. A bootstrap failure at seed 10002 changes to a stable reviewed success. Additional data and retraining have not been isolated using a retraining-only control. The already-inspected held-out seeds are appropriate for a debugging before/after comparison, while final method selection requires a separate untouched evaluation set.

Bootstrap measured correction

Fast measured correction

13 / FAILURE CASE AND CONDITIONING

The persistent pregrasp stall

Observed behavior is specific; the root cause remains unresolved.

Fast seed 10000	Baseline checkpoint	After feedback
Observed phase	pregrasp only	pregrasp only
Episode budget	300 steps / 15 seconds	300 steps / 15 seconds
Final recorded pregrasp EEF error	66.22 mm	55.30 mm
Advance threshold	28 mm	28 mm
Maximum can lift	0.062 mm	0.062 mm
Grasp contact	Not observed	Not observed
Final can-goal XY error	756.73 mm	756.73 mm
Arm collision evidence	No reported collision in review evidence	No reported collision in review evidence

The source-grounded teacher succeeds from the same initial features. Therefore the learned failure does not demonstrate an unreachable physical target. The policy approaches but settles outside the phase transition threshold, so no subsequent grasp/lift sequence begins. Both checkpoints show this symptom.

Two testable hypotheses, not established causes

Conditioning mismatch: five teacher-specific grasp/clearance settings are pooled, but the policy input does not explicitly include those parameters or the waypoint. Evaluation uses one fixed parameter set.
Distribution shift: initial object_x and goal_minus_object_x are outside the full training feature range in the reviewed failure; the largest initial absolute z-score is 7.55 on object_minus_eef_y. Range/z-score observations alone do not establish causality.

Next controlled comparison
Compare fixed-parameter teachers, parameter-conditioned BC, and the existing mixed-teacher baseline at equal training budget. Log waypoint error, XYZ outputs, normalized features, and phase transitions over time. Keep seed 10000 as a regression case; evaluate final gains on new untouched seeds and multiple training seeds.

Numeric failure and conditioning evidence

14 / OPEN ISSUES

Nine explicit limits and acceptance gates

Observed symptoms, confirmed implementation structure, and untested branches remain separate.

ID / priority	Current problem	Evidence status / next acceptance gate
P01 / P0	Learned policy stalls before grasp.	Observed at fast seed 10000. Require phase-completion and final-placement gains on new seeds without regressing current successes.
P02 / P0	Retargeting transfers grasp/lift summaries only; source benefit not isolated.	Structure confirmed. Compare source on/off and mismatched source at identical target starts/controller settings.
P03 / P0	Real human-video selection and task extraction untested.	No clip. Require actual frames, scores, Qwen raw text, reviewed JSON, and downstream use.
P04 / P1	Generic feedback does not improve fast success.	9/10 stays 9/10; mean error worsens. Add targeted-data and retraining-only controls.
P05 / P1	Historical failure code overstates reachability evidence.	Rename future observed timeout code; preserve old CSV with migration history.
P06 / P1	Teacher conditioning parameters absent from BC input.	Structure confirmed; causal effect unproven. Compare fixed versus explicit conditioning.
P07 / P1	Small evaluation and privileged/manual state dependencies.	Expand training/evaluation seeds and state estimation only when making broader claims.
P08 / P1	Default videos show only the first evaluation seed.	Review media adds success/failure galleries and labels. Integrate representative export into future pipeline runs.
P09 / P1	Default native success ends without stability dwell.	Review adds specific-Can/release/one-second checks. Integrate as additional fields without rewriting old metrics.

P0 items define the current research claim and main unsolved failure. P1 items must be addressed or explicitly held fixed before a stronger comparison. The review does not relabel absent experiments as completed fixes.

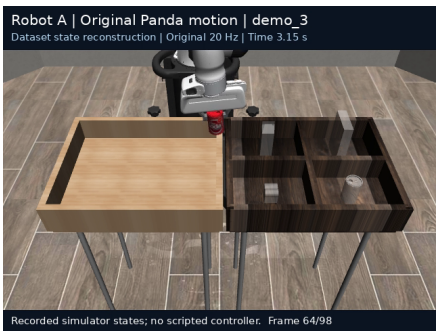
Original issue register and evidence

English stage-specific limitations

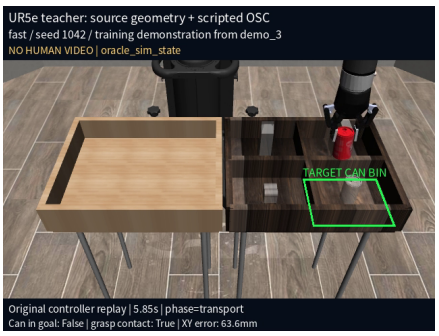
15 / THREE CASE STUDIES

Source lineage is many-to-one

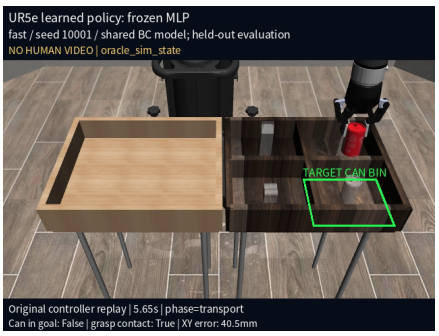
Three visual case studies use three source demonstrations, but their B videos are executions of the same pooled-data baseline policy.



Source A: demo_3
Original recorded states



Target B teacher: 1042
Source-grounded controller



Learned B: 10001
Shared baseline checkpoint

Illustrative frames occur at different times; they do not imply synchronized full-motion transfer. Video identities, frame indices, and hashes are preserved with the report figures.

Case	Source A demo	Executed training-teacher lineage	B held-out example
01	demo_3	UR5e seeds 1042 and 1047; source-derived grasp/lift parameters.	Seed 10001; 150 steps; 8.48 mm terminal XY error.
02	demo_5	UR5e seeds 1043 and 1048; source-derived grasp/lift parameters.	Seed 10003; 156 steps; 4.86 mm terminal XY error.
03	demo_7	UR5e seeds 1044 and 1049; source-derived grasp/lift parameters.	Seed 10009; 225 steps; 10.33 mm terminal XY error.

Do not read a case row as an individually trained A-to-B model
The shared baseline B checkpoint was trained on **all five** selected source IDs: demo_3, demo_5, demo_7, demo_4, and demo_1, through ten executed UR5e teacher episodes. Each case shows a source example and a successful B policy execution. The case pairing does not establish that one source alone caused that B behavior.

The three held-out B seeds are absent from the training-teacher set. Later video replays use frozen checkpoint weights and do not retrain. The reviewed successful B examples also remain inside the assigned Can compartment with no grasp contact after an additional second under the same learned policy.

Case 01: demo_3 and shared-policy B execution

Case 02: demo_5 and shared-policy B execution

Case 03: demo_7 and shared-policy B execution

Training lineage table

Shared checkpoint and case manifest

16 / SOURCE VIDEO PROVENANCE

Recorded states versus executed actions

The new source A videos reconstruct original HDF5 states. Earlier Panda videos show a separate live scripted policy.

Evidence type	How generated	What it establishes
Original-source state reconstruction	Restore each demo's original model XML using compatible installed asset paths. Set recorded state vector [time(1), qpos(37), qvel(33)] and forward kinematics for each frame. No physics step advances a source action.	Visualizes the recorded demonstration trajectory and supports source-state/observation alignment checks.
Live scripted Panda source branch	Execute hand-coded waypoint/gripper actions using current simulator feedback.	Shows the source-policy branch can run in the current simulator; it is not a replay of the original HDF5 demonstration.
Executed UR5e source-grounded teacher	Run current target OSC feedback control using extracted source grasp/lift parameters.	Provides actual target action/state trajectories for successful-teacher BC training.
Learned B video	Execute the frozen shared baseline policy on a held-out target seed.	Provides an independent target rollout; it does not replay teacher actions.

The source HDF5 records robosuite 1.5.1, while the current simulator is 1.5.2. Reusing original XML and numerically checking reconstructed Can/EEF positions is therefore an explicit compatibility step. A successful state reconstruction does **not** prove that replaying the recorded actions under newer physics reproduces the same dynamics or task success.

Source	Recorded duration	Max position error (m)	Evidence
demo_3	98 frames 4.90 s at 20 Hz	Can: 0.0 EEF: 4.81e-16 Tolerance: 1e-08	All-frame source metadata
demo_5	134 frames 6.70 s at 20 Hz	Can: 0.0 EEF: 6.89e-16 Tolerance: 1e-08	All-frame source metadata
demo_7	115 frames 5.75 s at 20 Hz	Can: 0.0 EEF: 5.82e-16 Tolerance: 1e-08	All-frame source metadata

All recorded source frames pass the 1e-8 m position tolerance. Source clips visualize stored successful demonstrations (terminal reward=1, done=1 for demo_3/5/7); they do not validate execution of the old actions under new dynamics.

Original-state source videos, per-frame CSVs, and validation metadata

17 / REPRODUCIBLE EXECUTION

SSH-safe runs and exact artifacts

The runnable baseline is detached; report and media generation do not replace actual training/evaluation evidence.

```
cd /home/dheo/Cross_Robot_Real2Sim
bash scripts/setup.sh
bash scripts/preflight.sh
bash scripts/launch.sh bootstrap
bash scripts/status.sh RUN_ID
tail -f runs/RUN_ID/logs/pipeline.log
bash scripts/resume.sh bootstrap RUN_ID
bash scripts/stop.sh RUN_ID
```

launch.sh creates a detached tmux session with absolute project paths, activates the project environment inside it, exports CUDA_VISIBLE_DEVICES=0 and EGL settings, and stores run/session/PID metadata. Logs include timestamps. A project GPU lock prevents unintended concurrent project launches. tmux survives SSH/VS Code disconnection while the host stays running; it is not host-reboot persistence.

The supervisor uses pipefail so tee does not hide a failed pipeline, records an exit code, handles signals, and preserves artifacts for resumption. Status checks both live process/session identity and persisted outcomes; disappearance alone never means success. Resume validates frozen configuration, input/artifact hashes, JSON/NPZ/checkpoint readability, and seed splits.

Dependency and artifact reproduction

```
bash scripts/launch.sh fast
# Real human input later:
cp /path/to/real_human_can_clip.mp4 data/human/
bash scripts/setup.sh full
bash scripts/launch.sh full
# Rebuild this PDF, without retraining:
.venv/bin/python -m pip install -r requirements-report.txt
.venv/bin/python scripts/build_english_report.py
```

Simulation, optional perception, review plotting, and PDF dependencies have separate requirement files. Original environment snapshots and lockfiles remain historical evidence. Raw runs/checkpoints/datasets stay local and gitignored; compact metrics, manifests, selected videos, and this report are versioned.

Setup and detached runtime scripts

Exact bootstrap dependency lock

Report builder and editable source

18 / VERIFICATION AND RESEARCH NEXT STEPS

What to trust, what to test next

The evidence supports an honest baseline and specific next experiments, not an already-complete general real-to-sim method.

Verification area	Executed evidence / remaining boundary
Input and schema tests	HDF5 14D object-layout regression; task validation; unknown physics; missing-human behavior; artifact contracts.
Simulator and controller tests	Headless Panda/UR5e steps and render; native goal predicate; action dimension/clipping; seeded reset and frame transforms.
Learning and diagnosis tests	Shared online 26D feature contract; successful-teacher filtering; held-out seed separation; saved-policy normalization/load; optimizer/sampling-state resume; actual-state diagnostic cases.
Runtime and integration tests	Isolated tmux process fixtures verify detached launch, duplicate-run prevention, nonzero exit preservation, scoped stop; missing-human full pipeline skips training.
Report QA	Embedded DejaVu fonts, English text extraction, page count, text margin bounds, layout audit, and rendered contact sheets. Report QA is not a new robotics experiment.

A prioritized experimental sequence

First: resolve the pregrasp stall with equal-budget fixed-parameter versus parameter-conditioned policies, plus a retraining-only control.
Second: isolate source-information value with source-on/source-off/mismatched-source ablations under matched target starts.
Third: validate real human input end to end and only then expand evaluation seeds, training seeds, state-estimation conditions, and physical-robot claims.

Primary references and evidence entry points

Robomimic Can PH v1.5 dataset
Robosuite controller documentation
CLIP model card (planned full branch)
Qwen2.5-VL-3B model card (planned full branch)

Experiment index and original reports

English source/B videos and lineage

Numerical failure evidence

PDF layout and font QA

All measurements in this edition are traced to preserved execution or review artifacts. No real human clip, full-model inference, source-action equivalence, unseen-task transfer, or physical-robot execution is claimed.